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Episode 47 · November 4, 2026 · 1 paper

Information Design for Behaviorally Stable Platooning

SKIM

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An operator who knows more about corridor fuel savings can raise how many trucks voluntarily join platoons by choosing what to announce publicly, but only if the trucks settle on the good equilibrium.

PROBLEM Voluntary platooning can fail at the decision to join: each truck pays a private cost to commit, savings depend on uncertain corridor conditions and on whether partners show up, and so no one joining is always an equilibrium. Prior platooning work mostly assumes participation and studies formation or control, so whether information alone can move participation had not been worked out.

METHOD They build a finite-vehicle coordination game that ties fuel savings to joining decisions, and they treat the operator's public announcement as a Bayesian persuasion design problem. They solve two cases. When joining costs are private, vehicles respond only to the posterior mean. When each vehicle privately observes the corridor state, the operator needs richer posterior information and formation outcomes become correlated. They give computational methods for the optimal signal in each case, focusing on two-message disclosure rules.

RESULTS In one literature-anchored reference scenario, optimized two-message disclosure supports 8.74 percent participation with private costs and 51.83 percent with private observations. That compares with zero under no disclosure, and 5.15 and 39.27 percent under full disclosure. Gains concentrate at intermediate opportunity levels, depend on how joining costs are calibrated, and do not necessarily maximize welfare.

CAVEAT Every gain assumes favorable symmetric equilibrium selection, and the authors themselves concede that public information cannot remove the no-participation equilibrium, so the headline numbers describe a best case, not a prediction. The evidence is one calibrated scenario plus sensitivity sweeps. The private-cost result of under nine percent participation is small in absolute terms. The operator is assumed to be informed, truthful and able to commit to a disclosure rule. And the physical side is reduced to a fuel-savings function, with no communication, joining maneuver or string stability modelled.

WHY IT MATTERS This touches your platoon formation interest, but at the level of economic incentives, not coordination protocols. There is no vehicle-to-vehicle message exchange, no proposal and acceptance step, and no control or communication constraint. The one transferable idea is that a costly commitment without a guaranteed partner creates a coordination trap that broadcast information cannot fix. That argues for explicit pairing or acceptance handshakes in a joining protocol, which is your territory. Otherwise it is peripheral.

arXiv:2611.00584v1 PDF incremental